



Original Investigation | Equity, Diversity, and Inclusion

Breast Cancer Incidence in Asian American, Native Hawaiian, and Pacific Islander Populations, 2000-2022

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Abstract

IMPORTANCE Breast cancer incidence among Asian American, Native Hawaiian, and Pacific Islander females as an aggregated group have been increasing rapidly. The extent to which these trends apply across Asian American, Native Hawaiian, and Pacific Islander ethnic groups is unclear.

OBJECTIVE To examine incidence trends of invasive breast cancer in 7 Asian American (Asian Indian or Pakistani, Chinese, Filipino, Japanese, Korean, Laotian or Kampuchean, and Vietnamese) and 2 Native Hawaiian or Pacific Islander (Guamanian, Chamorro, and Samoan and Native Hawaiian) ethnic groups overall and by age, stage, and subtype.

DESIGN, SETTING, AND PARTICIPANTS This population-based, descriptive, cross-sectional study used National Cancer Institute Surveillance, Epidemiology, and End Results Program data contributed by 14 US states (3 in the Northeast, 2 in the Midwest, 4 in the South, and 5 in the West). Asian American, Native Hawaiian, and Pacific Islander females of any age diagnosed with invasive breast cancer between January 1, 2000, and December 31, 2022, were included. Data were analyzed between September 2025 and March 2026.

MAIN OUTCOMES AND MEASURES Annual (or triannual) percentage change (APC) and 95% CIs of incidence rates were estimated using joinpoint regression, by age (<50 years, ≥50 years), stage (localized, regional, distant), and subtype (hormone receptor [HR] and *ERBB2* [formerly *HER2*/neu] negativity and positivity).

RESULTS A total of 148 608 Asian American, Native Hawaiian, and Pacific Islander females with breast cancer (44 234 aged <50 years [29.8%] and 104 374 aged ≥50 years [70.2%] at diagnosis; 138 808 of Asian American [93.4%] and 9800 of Native Hawaiian or Pacific Islander [6.6%] race and ethnicity) were included, among whom 63.9% were diagnosed at a localized stage and 66.6% with the HR-positive and *ERBB2*-negative subtype. Increased incidence rates of invasive breast cancer were observed among Asian American females (APC, 2.34%; 95% CI, 1.83%-3.68%) from 2012 to 2022, surpassing the trend in other racial and ethnic groups. A smaller steady increase was observed among Native Hawaiian or Pacific Islander females (APC, 0.84%; 95% CI, 0.48%-1.29%) from 2000 to 2022. Significant increases in overall and early-onset (ie, age <50 years) breast cancer were evident in all Asian American, Native Hawaiian, and Pacific Islander ethnic groups, with Chinese and Vietnamese females experiencing increased APCs of 4.57% (95% CI, 2.03%-8.36%) and 4.30% (95% CI, 1.98%-8.93%) since 2015 and 2016, respectively. Incidence increases were most pronounced for distant stage disease, with APCs of 4.02% (95% CI, 2.74%-5.72%) for Asian Indian and Pakistani females and 4.52% (95% CI, 2.55%-7.09%) for Chinese females. All Asian American groups, except Laotian and Kampuchean females, experienced increasing trends of HR-positive and *ERBB2*-negative cancer, with APCs ranging from 2.10% (95% CI, 0.74%-3.45%) for Japanese females

(continued)

Key Points

Question What are the trends in invasive breast cancer incidence rates in Asian American, Native Hawaiian, and Pacific Islander ethnic groups?

Findings In this cross-sectional study of 148 608 Asian American, Native Hawaiian, and Pacific Islander females, breast cancer incidence rates have been increasing at a rate of more than 3% per year in nearly all Asian American ethnic groups and has been especially pronounced for females aged less than 50 years, with advanced stages of disease, and with hormone receptor-positive and *ERBB2* (formerly *HER2*/neu)-negative and triple-negative subtypes.

Meaning These findings highlight that research and public health efforts among distinct Asian American, Native Hawaiian, and Pacific Islander ethnic groups are needed to identify contributing factors, promote awareness, and increase screening for breast cancer, especially given the rapid increases in early-onset cancer.

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Abstract (continued)

to 6.00% (95% CI, 3.62%-8.78%) for Korean females since 2010, whereas Native Hawaiian and Pacific Islander groups saw stable trends. Nearly all Asian American groups experienced increases in triple-negative breast cancer, with an APC as high as 6.17% (95% CI, 3.30%-11.73%) among Chinese females for 2017 to 2022.

CONCLUSIONS AND RELEVANCE This cross-sectional study found that breast cancer incidence rates increased rapidly across Asian American, Native Hawaiian, and Pacific Islander ethnic groups over a 20-year period. Research tailored to these distinct ethnic groups is needed to discern potentially novel risk factors for breast cancer. Culturally sensitive efforts are needed to promote awareness and increase breast cancer screening in distinct ethnic groups. The rapid recent increases in breast cancer incidence in Asian American, Native Hawaiian, and Pacific Islander women, especially early-onset disease, warrant urgent attention.

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Introduction

In the US, lower breast cancer incidence rates have historically been observed among Asian American females and higher rates among Native Hawaiian and Pacific Islander females compared with non-Hispanic White females.¹ This trend may be changing as rates over the 2 decades leading up to the mid-2010s were found to be increasing substantially in Asia^{2,3} and among distinct Asian American, Native Hawaiian, and Pacific Islander ethnic groups, as well as in aggregate, in the US.⁴⁻⁹ Data (1988-2009) from several Asian cancer registries (China, Hong Kong, South Korea, Taiwan, Singapore) have suggested that rapidly increasing trends were predominantly seen in recent and younger (aged 30-49 years) cohorts,^{3,10} with rates in these countries on pace to surpass the high rates among US non-Hispanic White females.^{3,10}

In the latest annual report to the nation, Asian American, Native Hawaiian, and Pacific Islander females had the steepest rise in breast cancer in all US racial and ethnic groups from 2017 to 2021 at 3.3% per year compared with 1.6% in all racial and ethnic groups combined.¹ Because the Asian American, Native Hawaiian, and Pacific Islander category comprises more than 40 ethnic groups with diverse cultures and immigration patterns, it is not surprising that breast cancer incidence varies across ethnic groups. Although all Asian American, Native Hawaiian, and Pacific Islander ethnic groups except Japanese have previously shown statistically increasing trends in breast cancer incidence,^{4,5,8} annual percentage change (APC) increases from 1990 to 2014 ranged from 0.65% in Chinese to 2.55% in Korean females.⁸ In Hawaii, breast cancer incidence in Japanese and Native Hawaiian females exceeded those of non-Hispanic White females,^{11,12} with high and increasing incidence among younger Japanese females and high incidence among older Native Hawaiian females.⁶ However, national breast cancer incidence trends for disaggregated Asian American, Native Hawaiian, and Pacific Islander ethnic groups have not been reported since 2014 due to the unavailability of annual population estimates for rate denominators.

Given the heterogeneity of breast cancer, examining trends by demographic and clinicopathologic characteristics could inform research on the underlying drivers and identify subpopulations for prevention. The rise in rates from 1990 to 2014 in Asian American ethnic groups was primarily driven by substantial increases in estrogen receptor-positive breast cancer rates, with APCs ranging from 1.21% in Chinese females to 3.46% in Korean females, while estrogen receptor-negative rates remained stable or decreased.⁸ Several studies have shown that Asian American, Native Hawaiian, and Pacific Islander women with breast cancer are more likely to be diagnosed with *ERBB2* (formerly *HER2*)/neu-positive breast cancers than other racial and ethnic groups.^{5,6,11,13-15} Asian American, Native Hawaiian, and Pacific Islander females aged 65 years or older experienced the fastest increase in incidence of triple-negative breast cancer (TNBC) at 5.5% from 2016 to 2019.¹⁶

More recently, there have been concerns regarding a predominance of early-onset breast cancer among Asian American, Native Hawaiian, and Pacific Islander females,^{13,14,17} with incidence rates at age less than 50 years increasing 4.0% annually from 2016 to 2019.¹⁶ Earlier data showed large increases in distant-stage disease among Asian American, Native Hawaiian, and Pacific Islander ethnic groups,^{5,11} indicating that observed incidence trend increases were not explained by increased screening.

The National Cancer Institute (NCI) Surveillance, Epidemiology, and End Results (SEER) Program recently updated the Asian American, Native Hawaiian, and Pacific Islander incidence trends database with ethnic group-specific denominator data, enabling examination of national trends for disaggregated groups through 2022. Given the concerning emerging breast cancer incidence patterns seen for Asian American, Native Hawaiian, and Pacific Islander females as an aggregated group, we leveraged this database to provide updated trends for 7 Asian American and 2 Native Hawaiian and Pacific Islander groups overall and by age, stage, and subtype.

Methods

Cancer Incidence Data

This cross-sectional study used data from female invasive breast cancer cases diagnosed from January 1, 2000, through December 31, 2022, obtained from the NCI SEER Program based on the Incidence-SEER Research Plus Limited Field Specialized Data (with detailed Asian and Pacific Islander populations), 21 registries (excluding Alaska and Seattle, Washington), November 2024 Submission (2000-2022) database,¹⁸ which covers 14 states (3 in the Northeast, 2 in the Midwest, 4 in the South, and 5 in the West) and 66.4% of the Asian American, Native Hawaiian, and Pacific Islander population. As data are deidentified and publicly available, institutional review board review and informed consent were not required in accordance with the Common Rule. This study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline for cross-sectional studies.

Based on American Community Survey data, the Asian American, Native Hawaiian, and Pacific Islander populations in the 14 SEER states were similar with regard to educational attainment and nativity to the Asian American, Native Hawaiian, and Pacific Islander population in non-SEER states, although with a slightly higher proportion of US-born individuals in SEER vs non-SEER states at 37% and 34%, respectively, for Asian American and 76% and 73%, respectively, for Native Hawaiian and Pacific Islander groups. Detailed Asian American, Native Hawaiian, and Pacific Islander ethnicity information was enhanced using the North American Association of Central Cancer Registries Asian Pacific Islander identification algorithm for assigning cases classified as Other Asian and Asian Not Otherwise Specified and Other Pacific Islander and Pacific Islander Not Otherwise Specified to more specific Asian American, Native Hawaiian, and Pacific Islander ethnic groups based on birthplace and names.^{19,20}

Although the focus of this report is on detailed Asian American, Native Hawaiian, and Pacific Islander ethnicities, we also assessed trends for non-Hispanic American Indian or Alaska Native, Hispanic, non-Hispanic Black, and non-Hispanic White populations for comparison. The North American Association of Central Cancer Registries Hispanic identification algorithm was used to improve accuracy of Hispanic ethnicity based on birthplace, race, and surname.²¹

Asian American, Native Hawaiian, and Pacific Islander were disaggregated into the following groups: (1) all Asian Americans; (2) Asian Indian and Pakistani (combined in SEER data); (3) Chinese; (4) Filipina; (5) Japanese; (6) Korean; (7) Laotian and Kampuchean (combined in analysis); (8) Vietnamese; (9) all Native Hawaiian and Pacific Islander; (10) Native Hawaiian, and (11) Guamanian, Chamorro, and Samoan (combined in analysis). We present case distributions for the 2 groups not otherwise specified (eTable 1 in Supplement 1) but did not calculate incidence rates as population estimates were not defined the same as cases. The Other Asian and Asian Not Otherwise Specified group represented 8.1% of all Asian American patients, and the Other Pacific Islander and Pacific

Islander Not Otherwise Specified group represented 22.5% of all Native Hawaiian and Pacific Islander patients.

We used the SEER summary stage variable (localized, regional, distant), available for all 21 registries for 2004 to 2022. Molecular subtype information was available beginning in 2010. Trends were assessed for 4 subtypes of breast cancer: hormone receptor (HR) (estrogen and progesterone receptors) positive and *ERBB2* negative, HR negative and *ERBB2* positive, HR positive and *ERBB2* positive, and HR negative and *ERBB2* negative (TNBC). Due to small counts for some groups, 3-year cumulative age-adjusted rates were used to assess trends. Incidence rates were not computed for unknown stage and subtype.

Population Data

For this analysis, annual state-level population estimates and disaggregated Asian American, Native Hawaiian, and Pacific Islander ethnic groups for 2000 to 2022 were provided by Woods & Poole Economics, Inc,²² with all populations stratified by sex (female, male) and 20 age groups (0, 1-4... 85-89, ≥ 90 years). These populations were derived from demographic survival models using 2000, 2010, and 2020 census data. The Native Hawaiian populations included all females of Hawaiian descent, encompassing the entirety of multiracial Hawaiians. All other specific Asian American, Native Hawaiian, and Pacific Islander groups were derived based on the bridged-race definition, which assumes that a portion of multiracial individuals identified themselves as that race. These customizations ensured comparability between the population and incidence data.²³ These populations are also consistent with NCI's Vintage 2023 county-level bridged-race populations²⁴ when counties are aggregated to states and Asian American, Native Hawaiian, and Pacific Islander groups are aggregated to a single group.

Statistical Analysis

The data analysis was performed between September 2025 and March 2026. Using SEER*Stat software, version 9.0.41.4 (NCI), incidence rates and 95% CIs (Tiwari modification) were calculated per 100 000 females and age adjusted to the 2000 US standard population. Annual or triannual percentage change in rates and 95% CIs (empirical quantile method) were calculated using the Joinpoint regression program, version 5.4.0.0 (NCI) and Excel, version 2021 (Microsoft Corp). Joinpoints indicate statistically meaningful changes in trends. The number of allowable joinpoints (omitting data from 2020, except for triannual trends based on recommended guidelines²⁵) was based on the number of data points and determined from the weighted bayesian information criterion as follows: 4 joinpoints for trends by age group (2000-2022), 3 joinpoints for stage (2004-2022), and 2 joinpoints for subtype (2010-2022). We excluded the earliest 2 years (2000-2001) for assessing trends by age, 1 year (2004) for stage, and 1 year (2010) for subtype when calculating triannual trends. The joinpoint method inherently reduces sensitivity to extreme observations by localizing their influence within segments and downweighting less precise estimates, thereby improving robustness of estimated APCs. Results were considered significant if the 95% CI did not overlap 0.

Results

A total of 148 608 Asian American, Native Hawaiian, and Pacific Islander females with invasive breast cancer were included (44 234 aged <50 years [29.8%] and 104 374 aged ≥ 50 years [70.2%] at diagnosis; 138 808 of Asian American [93.4] [16.9% Asian Indian or Pakistani, 23.9% Chinese, 25.2% Filipina, 11.4% Japanese, 7.3% Korean, 0.8% Laotian or Kampuchean, 0.6% Vietnamese, and 8.1% Other Asian American or Asian Not Otherwise Specified] and 9800 of Native Hawaiian and Pacific Islander [6.6%] [66.8% Native Hawaiian; 10.7% Guamanian, Chamorro, or Samoan; 22.5% Other Pacific Islander and Pacific Islander Not Otherwise Specified] race and ethnicity; 63.9% diagnosed at a localized stage; 66.6% diagnosed with HR-positive and *ERBB2*-negative subtype). The distributions

of breast cancer cases by selected demographic and clinicopathologic characteristics by Asian American, Native Hawaiian, and Pacific Islander ethnic groups are shown in eTable 1 in [Supplement 1](#) and for aggregated racial and ethnic groups, in eTable 2 in [Supplement 1](#). The highest proportion of cases were diagnosed before age 50 years in Asian Indian and Pakistani (35.5%), Korean (35.5%), Laotian and Kampuchean (39.8%), Vietnamese (36.7%), and Other Asian or Asian Not Otherwise Specified (36.5%) females. The groups with the lowest proportion of cases diagnosed at the localized stage were Asian Indian and Pakistani (59.6%), Laotian and Kampuchean (54.0%), and Pacific Islander (54.3%) females. Japanese (74.4%) and Native Hawaiian (74.4%) females had the highest proportion of HR-positive and *ERBB2*-negative diagnoses; Filipina (18.7%), Laotian and Kampuchean (21.4%), and Vietnamese (19.8%) females had the highest proportion of *ERBB2*-positive diagnoses, and Asian Indian and Pakistani (10.8%), Korean (10.1%), and Laotian and Kampuchean females (12.5%) had the highest proportion of TNBC.

The **Table** shows the incidence rates for 2018 to 2022 for Asian American, Native Hawaiian, and Pacific Islander ethnic groups, with rates for aggregated racial and ethnic groups shown in eTable 3 in [Supplement 1](#). Incidence rates per 100 000 varied substantially across Asian American, Native Hawaiian, and Pacific Islander ethnic groups, ranging from 54.1 (95% CI, 48.7-59.9) among Laotian or Kampuchean females to 177.2 (95% CI, 169.2-185.5) among Native Hawaiian females. Incidence rates per 100 000 for females aged less than 50 years ranged from 26.3 (95% CI, 22.0-31.1) among Laotian and Kampuchean females to 69.2 (95% CI, 64.4-74.4) among Japanese females. Distant stage rates per 100 000 were highest among Guamanian, Chamorro, and Samoan females (13.9; 95% CI, 9.7-19.2). Native Hawaiian and Japanese females had the highest rates per 100 000 of HR-positive and *ERBB2*-negative breast cancer at 131.7 (95% CI, 124.8-138.8) and 108.1 (95% CI, 103.9-112.5), respectively. The rates of TNBC per 100 000 were highest among Asian Indian and Pakistani (14.0; 95% CI, 13.1-14.9) and Japanese (13.7; 95% CI, 12.1-15.3) females.

Figure 1 and eTable 4 in [Supplement 1](#) show the joinpoint trends for Asian American, Native Hawaiian, and Pacific Islander ethnic groups, with trends for aggregated racial and ethnic groups shown in eFigure 1 in [Supplement 1](#). Asian American females experienced an increase in incidence from 2012 to 2022 (APC, 2.34%; 95% CI, 1.83%-3.68%), while Native Hawaiian and Pacific Islander females experienced a steady increase from 2000 to 2022 (APC, 0.84%; 95% CI, 0.48%-1.29%). Increases were seen for all Asian American, Native Hawaiian, and Pacific Islander ethnic groups except for Korean females, among whom the increases were not statistically significant. The largest increases were seen for Chinese females (APC, 4.39%; 95% CI, 2.58%-7.70%) from 2016 to 2022, Vietnamese females (APC, 4.12%; 95% CI, 2.32%-8.10%) from 2015 to 2022, and Asian Indian and Pakistani females (APC, 4.17%; 95% CI, 3.20%-7.81%) from 2008 to 2015, decreasing to 1.94% (95% CI, 0.42%-2.70%) thereafter.

Figure 2 and eTable 4 in [Supplement 1](#) show the trends across Asian American, Native Hawaiian, and Pacific Islander ethnic groups aged less than 50 years and 50 years or older, with trends for aggregated racial and ethnic groups shown in eFigure 2 in [Supplement 1](#). Among patients younger than 50 years at diagnosis, rates among Asian American, Native Hawaiian, and Pacific Islander females increased (APC, 2.88%; 95% CI, 1.59%-5.21%) for 2016 to 2022 to comparable rates as non-Hispanic White females in 2021 and 2022. Nearly all Asian American, Native Hawaiian, and Pacific Islander ethnic groups experienced statistically significant increases at age less than 50 years at diagnosis, with the largest increases seen for Chinese females (APC, 4.57%; 95% CI, 2.03%-8.36%) from 2017 to 2022 and Vietnamese females (APC, 4.30%; 95% CI, 1.98%-8.93%) from 2016 to 2022. The increases were also stark for Asian Indian and Pakistani females (APC, 2.09%; 95% CI, 1.65%-2.72%) and Japanese females (APC, 2.41%; 95% CI, 1.77%-3.14%) from 2000 to 2022, Korean females (APC, 2.86%; 95% CI, 2.25%-3.62%) from 2000 to 2022, and Native Hawaiian females (APC, 1.14%; 95% CI, 0.38%-2.02%) from 2000 to 2022.

Asian American, Native Hawaiian, and Pacific Islander females aged 50 years or older at diagnosis also experienced the most rapid increase in incidence rates of any racial and ethnic group (APC, 2.52%; 95% CI, 1.62%-3.91%) since 2016. All Asian American, Native Hawaiian, and Pacific

Table. Age-Adjusted Incidence Rates for Females With Breast Cancer by Asian American, Native Hawaiian, and Pacific Islander Ethnic Group, 2018-2022, SEER-21 Program											
Characteristic	Incidence per 100 000 women (95% CI)										
	Asian American	Asian Indian and Pakistani	Chinese	Filipina	Japanese	Korean	Laotian and Kampuchean	Vietnamese	Native Hawaiian and Pacific Islander	Native Hawaiian	Guamanian, Chamorro, and Samoan
Overall	110.5 (109.5-111.6)	133.6 (130.8-136.5)	106.0 (104.0-108.1)	119.5 (117.2-122.0)	145.8 (140.8-151.0)	94.5 (91.3-97.8)	54.1 (48.7-59.9)	81.8 (78.9-84.8)	169.1 (163.0-175.5)	177.2 (169.2-185.5)	100.9 (89.2-113.7)
Age at diagnosis, y											
<40	13.8 (13.3-14.3)	13.8 (12.9-14.8)	13.8 (12.8-14.9)	12.9 (11.7-14.1)	18.4 (15.4-21.9)	11.8 (10.2-13.6)	10.6 (7.9-14.0)	11.9 (10.4-13.6)	17.5 (15.1-20.2)	18.2 (15.0-21.9)	NR ^a
40-49	177.1 (173.5-180.7)	195.1 (177.5-193.1)	182.8 (175.2-190.6)	165.2 (157.4-173.4)	257.2 (237.6-277.9)	180.1 (168.2-192.7)	84.2 (67.2-104.3)	147.6 (137.6-158.2)	208.5 (190.4-227.9)	218 (193.8-244.4)	89.3 (63.2-122.6)
50-59	241.5 (236.9-246.0)	298.0 (285.6-310.8)	236.3 (227.5-245.3)	251.6 (241.4-262.0)	284.0 (263.4-305.6)	213.1 (198.8-228.1)	136.6 (112.1-164.9)	173.6 (161.8-186.1)	337.9 (313.6-363.6)	323.6 (293.2-356.3)	213.4 (168.8-266.2)
60-69	300.8 (295.3-306.4)	399.2 (382.1-416.8)	269.2 (259.1-279.5)	365.9 (353.0-379.1)	385.0 (360.6-410.7)	238.1 (221.8-255.2)	147.9 (119-181.6)	216.5 (201.7-232.1)	524.7 (489.8-561.4)	554.1 (509.2-601.9)	352.7 (282.8-434.8)
70-79	319.9 (312.5-327.4)	422.3 (399.3-446.3)	284.7 (270.8-299.1)	390.4 (373.5-407.9)	428.3 (399.1-459.2)	224.6 (205.2-245.5)	116.3 (81.5-161.3)	207.8 (188.1-229.0)	623.0 (571.6-677.9)	682.9 (616.7-754.2)	432.4 (324.4-565.4)
≥80	233.8 (225.4-242.5)	304.0 (273.9-336.6)	221.4 (205.9-237.8)	273.7 (253.7-294.9)	325.4 (297.6-354.9)	172.2 (149.3-197.6)	NR ^a	149.1 (125.2-176.1)	436.5 (375.3-504.6)	500.5 (420.4-591.1)	NR ^a
0-49	48.5 (47.7-49.4)	50.3 (48.5-52.1)	49.7 (47.9-51.6)	45.3 (43.4-47.3)	69.2 (64.4-74.4)	47.6 (44.8-50.6)	26.3 (22-31.1)	40.8 (38.3-43.4)	58.1 (53.8-62.7)	60.7 (54.9-66.9)	24.5 (18.3-32.0)
≥50	272.9 (269.9-275.9)	351.9 (342.7-361.2)	253.5 (247.8-259.2)	314.0 (307.1-321.0)	346.4 (333.6-359.5)	217.2 (208.4-226.3)	126.8 (111.1-144.3)	189.2 (181.2-197.6)	459.8 (440.8-479.5)	482.3 (457.8-507.7)	301.0 (262.4-344.0)
Stage											
Localized	71.7 (70.9-72.6)	84.2 (81.9-86.5)	70.8 (69.2-72.5)	75.7 (73.9-77.6)	104.8 (100.6-109.1)	62.3 (59.7-65.0)	29.5 (25.5-33.9)	51.1 (48.9-53.5)	105.1 (100.3-110.1)	116.3 (109.8-123.0)	52.2 (43.8-61.6)
Regional	30.1 (29.6-30.7)	37.7 (36.2-39.2)	27.3 (26.3-28.4)	33.9 (32.7-35.2)	33.1 (30.6-35.7)	26.1 (24.4-27.9)	17.1 (14.2-20.5)	23.5 (22.0-25.1)	47.9 (44.6-51.3)	47.7 (43.6-52.0)	31.9 (25.7-39.1)
Distant	5.9 (5.6-6.1)	8.4 (7.7-9.2)	5.1 (4.7-5.6)	7.5 (6.9-8.1)	5.3 (4.4-6.4)	3.8 (3.2-4.5)	4.9 (3.4-6.9)	3.8 (3.2-4.5)	12.8 (11.1-14.6)	10.0 (8.2-12.2)	13.9 (9.7-19.2)
Subtype											
HR ⁺ and ERBB2 ⁻ (formerly HER2/neu)	75.4 (74.6-76.3)	91.1 (88.7-93.5)	71.6 (70.0-73.4)	81.8 (79.9-83.8)	108.1 (103.9-112.5)	65.9 (63.2-68.6)	30.0 (26.0-34.5)	52.1 (49.8-54.4)	121.7 (116.5-127.1)	131.7 (124.8-138.8)	71.1 (61.3-82.0)
HR ⁻ and ERBB2 ⁺	6.2 (6.0-6.5)	6.3 (5.7-6.9)	6.3 (5.8-6.8)	7.9 (7.3-8.5)	6.2 (5.2-7.5)	4.5 (3.8-5.2)	4.9 (3.4-6.9)	5.0 (4.3-5.8)	8.0 (6.7-9.5)	8.9 (7.2-10.9)	NR ^a
HR ⁺ and ERBB2 ⁺	11.7 (11.4-12.1)	13.8 (12.9-14.7)	11.2 (10.5-11.9)	14.0 (13.2-14.8)	11.4 (10.0-13.1)	9.3 (8.3-10.4)	5.9 (4.3-8.1)	10.1 (9.1-11.2)	18.9 (16.9-21.1)	18.5 (16-21.3)	11.2 (7.8-15.6)
TNBC	9.9 (9.6-10.2)	14.0 (13.1-14.9)	8.9 (8.4-9.6)	8.7 (8.1-9.4)	13.7 (12.1-15.3)	8.9 (7.9-9.9)	6.7 (4.9-8.9)	7.5 (6.7-8.5)	12.5 (10.9-14.3)	11.5 (9.6-13.8)	8.6 (5.6-12.8)

Abbreviations: +, positive; -, negative; HR, hormone receptor; NR, not reported; SEER, Surveillance, Epidemiology, and End Results; TNBC, triple-negative breast cancer.

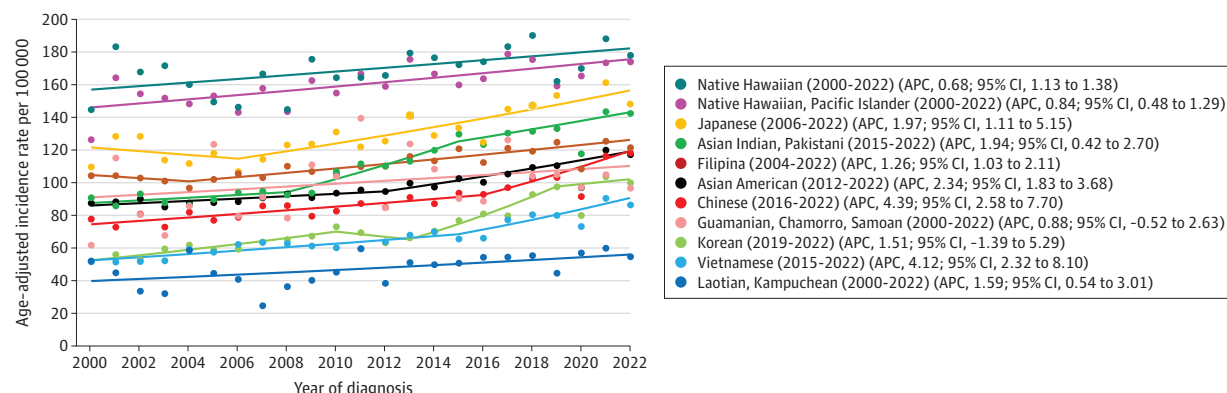
a Suppressed due to fewer than 16 cases.

Islander ethnic groups except for Native Hawaiian saw significantly increasing trends of breast cancer at age 50 years or older at diagnosis, with the largest increases for Korean females (APC, 5.45%; 95% CI, 4.00%-9.03%) after 2014, Chinese females (APC, 3.97%; 95% CI, 2.90-6.59) after 2014, and Asian Indian and Pakistani females (APC, 3.15%; 95% CI, 2.79%-4.40%) after 2004.

Trends by stage are shown in **Figure 3** and eTable 5 in **Supplement 1** for Asian American, Native Hawaiian, and Pacific Islander ethnic groups and in eFigure 3 in **Supplement 1** for aggregated racial and ethnic groups. From 2004 to 2021, Asian American, Native Hawaiian, and Pacific Islander females experienced increasing trends of breast cancer across all 3 stages, with the largest increase for distant stage (APC, 3.04%; 95% CI, 2.51%-3.71%). Most Asian American ethnic groups (but not Native Hawaiian and Pacific Islander groups) had statistically significant increases in distant-stage disease, including Asian Indian and Pakistani (APC, 4.02%; 95% CI, 2.74%-5.72%), Chinese (APC, 4.52%; 95% CI, 2.55%-7.09%), Filipina (APC, 2.33%; 95% CI, 1.45%-3.37%), and Korean (APC, 3.95% per 3 years; 95% CI, 0.01%-8.92% per 3 years) females.

Subtype trends are shown in **Figure 4** and eTable 6 in **Supplement 1** for Asian American, Native Hawaiian, and Pacific Islander ethnic groups and in eFigure 4 in **Supplement 1** for aggregated racial and ethnic groups. All Asian American groups, except for Laotian and Kampuchean, had significantly increasing trends of HR-positive and *ERBB2*-negative breast cancer, with the smallest increase seen for Japanese females (APC, 2.10%; 95% CI, 0.74%-3.45%) and the largest increases seen for Korean (APC, 6.00%; 95% CI, 3.62%-8.78%) and Vietnamese (APC, 5.01%; 95% CI, 3.78%-6.52%) females since 2010. Chinese females had increasing trends of HR-negative and *ERBB2*-positive breast cancer (APC, 11.47%; 95% CI, 2.21%-18.66%). Japanese and Native Hawaiian females also experienced large APC increases in HR-negative and *ERBB2*-positive breast cancer at 5.72% (95% CI, 2.22%-9.38%) from 2010 to 2022 and 5.07% per 3 years (95% CI, 0.50%-10.81% per 3 years) from 2011 to 2022, respectively. Incidence of HR-positive and *ERBB2*-positive breast cancer declined over time among Chinese females (APC, -3.21%; 95% CI, -6.68% to -0.80%) from 2017 to 2022 and Filipina females (APC, -5.97%; 95% CI, -8.79% to -3.09%) from 2019 to 2022, but increased among Japanese females (APC, 2.91%; 95% CI, 0.01%-5.70%) from 2010 to 2022. The incidence of TNBC increased at an APC of 4.35% (95% CI, 2.62%-8.27%) from 2016 to 2022 among Asian American, Native Hawaiian, and Pacific Islander females (largest of all racial and ethnic groups), with the largest increases for Asian Indian and Pakistani females (APC, 3.06%; 95% CI, 1.91%-4.42%) from 2010 to 2022, Chinese females (APC, 6.17%; 95% CI, 3.30%-11.73%) from 2017 to 2022, Filipina females (APC, 4.00%; 95% CI, 1.93%-9.44%) from 2016 to 2022, and Vietnamese females (APC, 3.86%; 95% CI, 1.38%-6.85%) from 2010 to 2022.

Figure 1. Line Graphs of Age-Adjusted Invasive Breast Cancer Incidence Rates and Trends by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity, All Ages, 2000-2022, Surveillance, Epidemiology, and End Results Program-21 Program



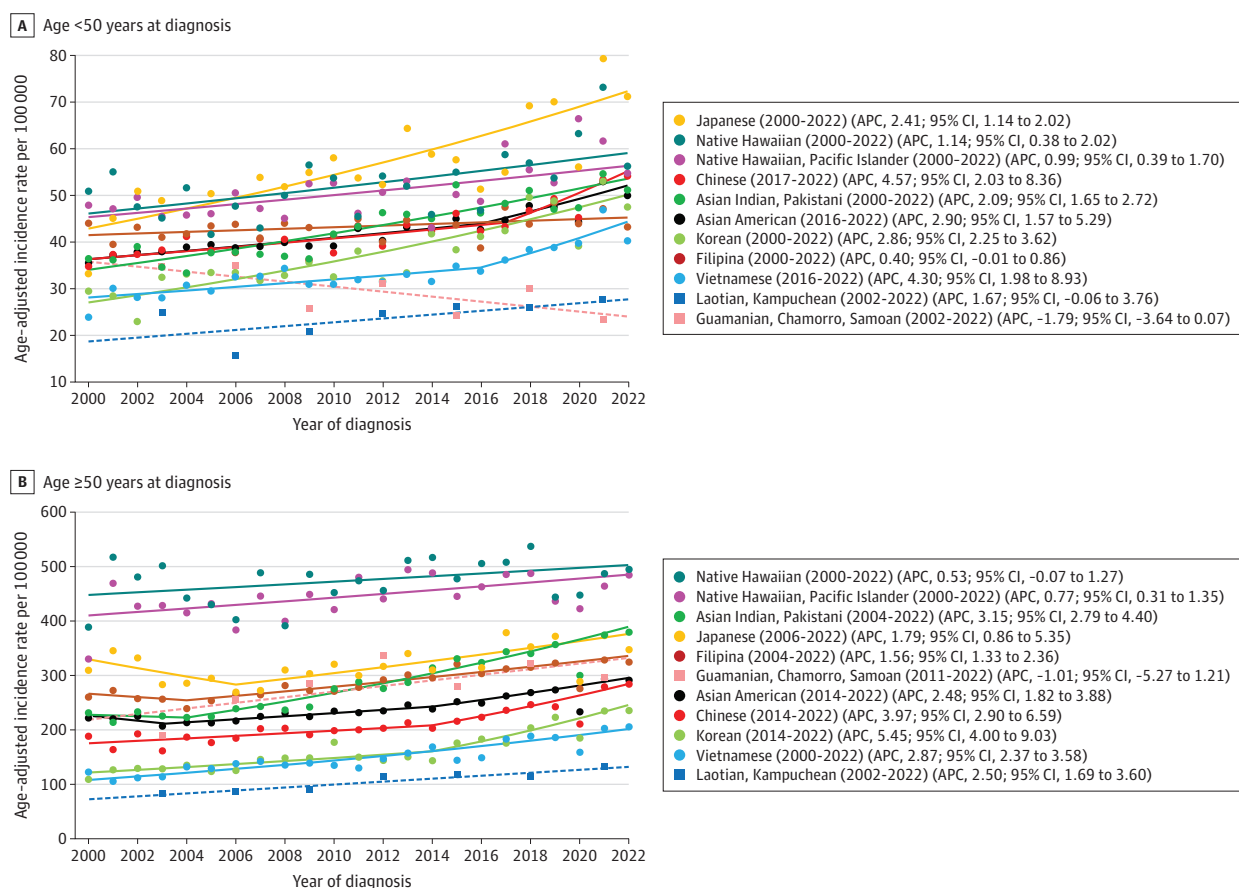
Annual percentage change (APC) and 95% CIs presented for the latest segment, generated by Joinpoint. Circles indicate annual age-adjusted incidence rates; lines, annual percentage change trends.

Discussion

Using the recently updated NCI SEER specialized database,¹⁸ we conducted a contemporary analysis of invasive breast cancer incidence trends among detailed Asian American, Native Hawaiian, and Pacific Islander populations for the 23-year period spanning 2000 to 2022. This cross-sectional study found significantly increasing rates of breast cancer among Asian American females overall from 2012 to 2022 at 2.34% per year, surpassing the increase in other racial and ethnic groups, and a steady increase among Native Hawaiian and Pacific Islander females overall at 0.84% per year from 2000 to 2022. The increase was particularly marked among Asian American females younger than 50 years at diagnosis at a rate of 2.90% per year. With this rapid increase, the early-onset incidence rate among Asian American, Native Hawaiian, and Pacific Islander females as an aggregate group is now comparable with the rate among non-Hispanic White females. Access to this special Asian American, Native Hawaiian, and Pacific Islander trends database is available upon request at the SEER website.

Similar to prior reports of data for 1990 to 2008,^{4,26} we found that breast cancer incidence rates varied widely across Asian American, Native Hawaiian, and Pacific Islander groups, with more than a 3-fold difference between the highest (177 per 100 000 among Native Hawaiian females) and lowest (54 per 100 000 among Laotian and Kampuchean females) rates. Variations in trends across Asian American, Native Hawaiian, and Pacific Islander ethnic groups were also evident, albeit

Figure 2. Line Graphs of Age-Adjusted Invasive Breast Cancer Incidence Rates and Trends by Detailed Asian American, Native Hawaiian, and Pacific Islander Race Ethnicity and Age at Diagnosis, 2000-2022, Surveillance, Epidemiology, and End Results Program-21 Program

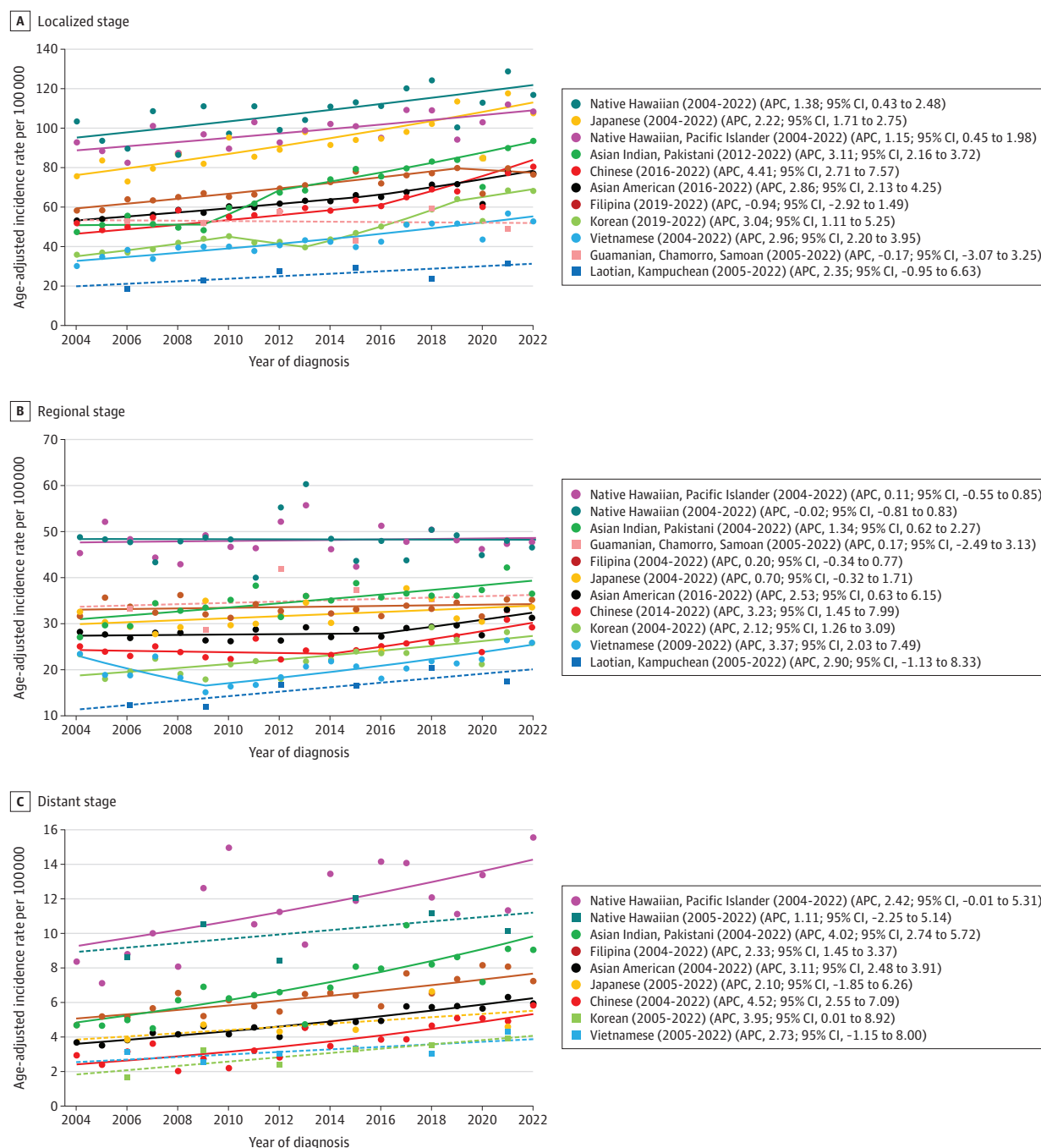


A, Annual (or triannual) percentage change (APC) and 95% CIs presented for the latest segment, generated by Joinpoint. B, Triannual percentage change (2002-2004, 2005-2007, 2008-2010, 2011-2013, 2014-2016, 2017-2019, and 2020-2022 [2000-2001

excluded]). Circles indicate annual age-adjusted rates; squares, 3-year cumulative rates; dotted line, linear trend generated by Excel; solid lines, linear trends generated by Joinpoint.

increases were seen for nearly all groups since 2000, with the largest increases for Chinese and Vietnamese women at more than 4% annually from 2015 and 2016 to 2022. Although incidence increases in older women were observed, particularly concerning is the rapidly increasing recent trends in early-onset breast cancer. These rapidly increasing breast cancer incidence rates mirror

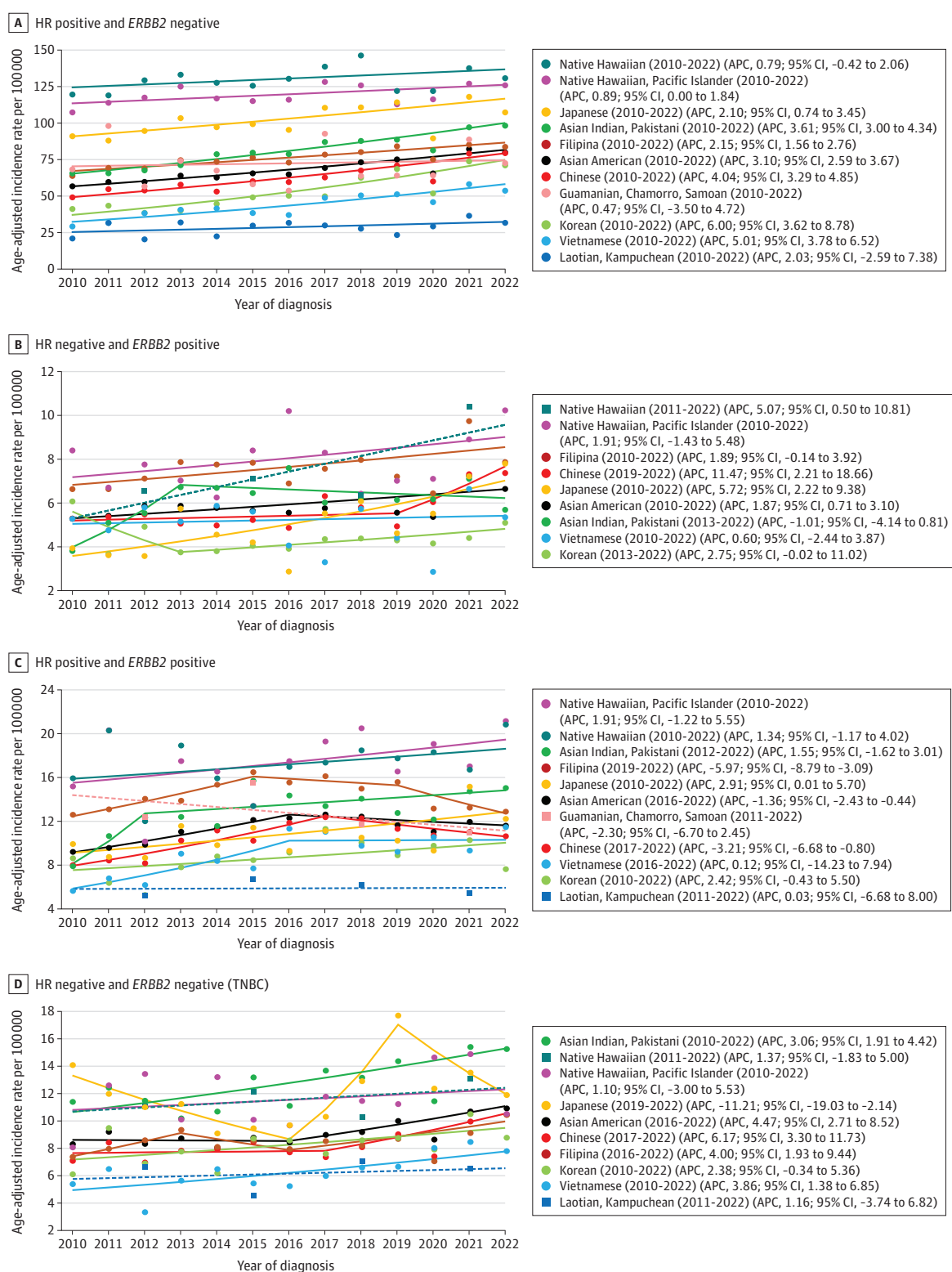
Figure 3. Line Graphs of Age-Adjusted Invasive Breast Cancer Incidence Rates and Trends by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity and Stage, 2004-2022, Surveillance, Epidemiology, and End Results Program-21 Program



A, Annual (or triannual) percentage change (APC) and 95% CIs presented for the latest segment, generated by Joinpoint. B-D, Triannual percentage change (2005-2007, 2008-2010, 2011-2013, 2014-2016, 2017-2019, 2020-2022 [excluding 2004]). Guamanian, Chamorro, and Samoan and Laotian and Kampuchean rates and triannual

percentage change suppressed due to fewer than 16 cases for at least 1 3-year period. Circles indicate annual age-adjusted rates; squares, 3-year cumulative rates; dotted lines linear trends generated by Excel; solid lines, linear trends generated by Joinpoint.

Figure 4. Line Graphs of Age-Adjusted Invasive Breast Cancer Incidence Rates and Trends by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity and Hormone Receptor (HR) and *ERBB2* (Formerly *HER2/Neu*) Subtype, 2010-2022, Surveillance, Epidemiology, and End Results Program-21 Program



A, Annual (annual or triannual in B-D) percentage change (APC) and 95% CIs presented for the latest segment, generated by Joinpoint. B-D, Triannual percentage change (2011-2013, 2014-2016, 2017-2019, 2020-2022 [excluding 2010]); Guamanian, Chamorro, and Samoan and Laotian and Kampuchean rates and triannual percentage change suppressed due to fewer than 16 cases for at least 13-year period. Circles indicate annual age-adjusted rates; squares indicate 3-year cumulative rates. Dotted linear trend lines generated by Excel; solid linear trend lines generated by Joinpoint. TNBC indicates triple-negative breast cancer.

trends in Asia,^{3,10,27} with these studies indicating that the increases may be more pronounced among recent and younger cohorts.^{3,10} The surge in incidence has been attributed to industrialization and adoption of western lifestyles, characterized by shifts in dietary habits and reproductive behaviors.¹⁴ Ecologic analyses correlating these incidence trends with risk factor trends may be informative, but population-level risk factor data for disaggregated Asian American, Native Hawaiian, and Pacific Islander ethnic groups are sparse. The recent trends in early-onset breast cancer among Asian American females may be seen primarily among recent birth cohorts^{28,29}; thus, we may continue to see a rising incidence in these cohorts with age. Two San Francisco Bay Area studies found that foreign-born Asian American, Native Hawaiian, and Pacific Islander females had significantly higher breast cancer risks than US-born Asian American, Native Hawaiian, and Pacific Islander females,^{30,31} with associations observed primarily in recent birth cohorts. Accounting for risk factors did not explain the patterns, suggesting novel risk factors. In addition, census data have shown that recent immigration cohorts from Asian countries tend to be of higher socioeconomic status than earlier cohorts.³² These emerging patterns highlight the need for research to understand the unique risk factors and biology of breast cancer in these populations. Given the lack of availability of population-level risk factor data for disaggregated Asian American, Native Hawaiian, and Pacific Islander groups, epidemiologic studies that capture these risk factors in relation to birth and immigration cohorts may be most fruitful to advance understanding of contributing etiologic factors.

The strikingly high and persistently stable incidence rate of breast cancer among Native Hawaiian and Pacific Islander females warrants attention. Notably, the incidence rate of breast cancer among Native Hawaiian females was 1.2-fold to 3.3-fold higher compared with other racial and ethnic groups and consistent across age, stage, and subtype. The prevalence of established risk factors for breast cancer in Native Hawaiian females is lower than that of non-Hispanic White females, and an elevated incidence has been found in Native Hawaiian compared with non-Hispanic White females, even after adjusting for breast cancer risk factors.³³ The underlying factors for this higher burden among Native Hawaiian females may be attributed to early-life exposures, hormonal levels, cultural factors, and structural and social drivers of health.

Increasing trends were observed across all stages of disease for many Asian American, Native Hawaiian, and Pacific Islander ethnic groups, with the most pronounced increases for distant stage, suggesting that the recent trends are not due to increased screening. Indeed, breast cancer screening rates and delayed follow-up of abnormal mammography results have been shown to be lower among some Asian American, Native Hawaiian, and Pacific Islander females compared with other racial and ethnic groups.^{34,35} From 2016-2018 National Health Information Survey data, up-to-date breast cancer screening among females aged 45 years or older were 55% among Asian Indian, 61% among Chinese, and 47% among other Asian females compared with 63% among non-Hispanic White females,³⁶ although up-to-date rates were higher for Asian American, Native Hawaiian, and Pacific Islander females as an aggregate in 2023.³⁷ At this time, more current screening data for disaggregated Asian American, Native Hawaiian, and Pacific Islander ethnic groups are not available. Health care access, insurance coverage, and cultural factors are strong drivers of mammographic screening among Asian American, Native Hawaiian, and Pacific Islander females.³⁸⁻⁴³ Implementation of effective interventions for these populations is urgently needed.³⁸ Continued monitoring by stage among younger Asian American, Native Hawaiian, and Pacific Islander females could be important to identify whether recent guidelines to start screening at earlier ages might reverse these trends.

We observed statistically significant recent increases in most Asian American, Native Hawaiian, and Pacific Islander ethnic groups for HR-positive and *ERBB2*-negative breast cancer and TNBC. Although HR-positive and *ERBB2*-negative breast cancer remains the most common subtype, the most rapid increases were seen for the more aggressive subtypes for some Asian American, Native Hawaiian, and Pacific Islander ethnic groups, including HR-negative and *ERBB2*-positive disease for Chinese, Japanese, and Native Hawaiian females and TNBC for Asian Indian and Pakistani, Chinese, Filipina, and Vietnamese females. The increases in HR-negative and *ERBB2*-positive breast cancer

and TNBC somewhat counter the earlier trends reported by Tuan et al,⁸ who showed that estrogen receptor-negative rates remained stable or decreased. Unknown subtype information was 14% among Asian American females and 9.3% among Native Hawaiian and Pacific Islander females in 2010, decreasing to 6.3% and 2.8%, respectively, in 2022. After 2016, subtype completeness was stable at 6% among Asian American females and 4% among Native Hawaiian and Pacific Islander females; thus, the significant incidence trends after 2016 may have been unaffected. The HR-negative and *ERBB2*-positive breast cancer subtype has known reproductive risk factors, such as age at menarche, menopause, and breastfeeding^{44,45}; however, associations may vary by race and ethnicity.⁴⁶ Further research is needed to identify relevant risk and protective factors for HR-negative and *ERBB2*-positive breast cancers in Asian American, Native Hawaiian, and Pacific Islander females.

Limitations

This study had several limitations. We noted an increasing number of breast cancer cases coded in the Other Asian or Asian Not Otherwise Specified category. Combining Other Asian and Asian Not Otherwise Specified into 1 category made it difficult to disentangle cases coded into defined Asian categories vs those without a specified ethnicity. It was also difficult to match this group to population estimate data; thus, incidence rates for the Other Asian and Asian Not Otherwise Specified and Other Pacific Islander and Pacific Islander Not Otherwise Specified groups should not be computed, as they have no corresponding population denominator. Given that nearly 10% of breast cancer cases in Asian American and more than 22% of cases in Native Hawaiian or Pacific Islander females were coded into these not otherwise specified categories, the estimated rates for the defined Asian American, Native Hawaiian, and Pacific Islander categories may be slightly underestimated. We also could not discount the possibility of race and ethnicity misclassification, particularly across Asian American, Native Hawaiian, and Pacific Islander groups. Prior research has shown minimal misclassification in specific Asian ethnic groups,⁴⁷⁻⁴⁹ but we acknowledge that more contemporary research is needed, especially given the increase in number and size of the smaller Asian ethnicities. In addition, the increase in the proportion of multiracial and multiethnic populations in the US creates challenges in ensuring comparability between numerator and denominator classifications. We urge cancer registries to make a concerted effort to work with reporting facilities to improve collection of specific Asian American, Native Hawaiian, and Pacific Islander ethnic categories. There may be errors in population denominator classifications as the annual estimates are based on statistical interpolations from multiple data sources. Although the 21 SEER registries cover 66% of the Asian American, Native Hawaiian, and Pacific Islander population, and they are demographically similar to those in non-SEER regions, our results may nonetheless not generalize to all Asian American, Native Hawaiian, and Pacific Islander individuals in the US. Finally, due to the COVID-19 pandemic, cancer incidence rates in 2020 were approximately 10% lower than rates in earlier years; prior research has shown a lack of a rebound effect (increased rates that incorporate potentially missed cases).⁵⁰⁻⁵³ In addition, we noted a proportional case shift to later stage at diagnosis in the years following the pandemic, returning to similar prepandemic proportions after the spring of 2021.⁵⁴ These studies indicate that as our trend evaluation extends through 2022, and with the exclusion of 2020 rates from our joinpoint analyses, the identified trends should have been minimally influenced by the pandemic. Nonetheless, our analyses follow standard recommended approaches and, thus, are comparable to other trend results covering the pandemic period.^{25,55}

Conclusions

This population-based, descriptive, cross-sectional study covering a more than 20-year period found rapid recent increases in breast cancer incidence in Asian American, Native Hawaiian, and Pacific Islander women, especially for early-onset breast cancer. Attention to mortality trends is needed over

the next several years to determine whether the rapid incidence increases, including for advanced-stage disease and aggressive subtypes, might translate into increasing mortality trends. In contrast to declining mortality trends in other racial and ethnic groups, mortality data through 2022 have indicated stable rates among Asian American, Native Hawaiian, and Pacific Islander females,⁵⁶ which may be due to a combination of increasing incidence rates overall and for late stage. Research specific to distinct Asian American, Native Hawaiian, and Pacific Islander ethnic groups is needed to discern potentially unique risk factors, which may inform risk factor discovery more broadly. Finally, linguistically and culturally tailored efforts that focus on the needs of specific Asian American, Native Hawaiian, and Pacific Islander communities may be required to overcome cultural stigma, promote awareness, and increase screening.

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Concept and design: Gomez, Yu, Shariff-Marco.

Acquisition, analysis, or interpretation of data: All authors.

Drafting of the manuscript: Gomez, McKinley, Chan, Yu, Scoppa, Gibson, Noone.

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SUPPLEMENT 1.

eTable 1. Demographic and Clinicopathologic Characteristics Among Females Diagnosed With Breast Cancer by Asian American, Native Hawaiian, Pacific Islander Detailed Ethnicity, 2018-2022, SEER-21

eTable 2. Demographic and Clinicopathologic Characteristics Among Females Diagnosed With Breast Cancer by Aggregated Racial and Ethnic Groups, 2000-2022, SEER-21

eTable 3. Age-Adjusted Incidence Rates and 95% CIs for Females Diagnosed With Breast Cancer by Aggregated Racial and Ethnic Group, 2018-2022, SEER-21

eTable 4. Trends in Breast Cancer Incidence Rates Among Females, by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity and Age Group at Diagnosis, 2000-2022, SEER-21

eTable 5. Trends in Breast Cancer Incidence Rates Among Females, by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity and Summary Stage at Diagnosis, 2004-2022, SEER-21

eTable 6. Trends in Breast Cancer Incidence Rates Among Females, by Detailed Asian American, Native Hawaiian, and Pacific Islander Ethnicity and Breast Cancer Subtype, 2010-2022, SEER-21

eFigure 1. Age-Adjusted Invasive Breast Cancer Incidence Rates and Joinpoint Trends by Aggregated Racial and Ethnic Groups, All Ages, 2000-2022, SEER-21

eFigure 2. Age-Adjusted Invasive Breast Cancer Incidence Rates and Joinpoint Trends by Aggregated Racial and Ethnic Groups, 2000-2022, SEER-21

eFigure 3. Age-Adjusted Invasive Breast Cancer Incidence Rates and Joinpoint Trends by Aggregated Racial and Ethnic Groups, by Stage, 2004-2022, SEER-21

eFigure 4. Age-Adjusted Invasive Breast Cancer Incidence Rates and Joinpoint Trends by Aggregated Racial and Ethnic Groups, by Subtype, 2010-2022, SEER-21

SUPPLEMENT 2.

Data Sharing Statement